

Temperature Distributions and Temperature-Standardized Ozone Change Across U.S. Monitoring Sites, 2015–2025

Ellis Varden

Independent Researcher

DOI: [10.5281/zenodo.21434897](https://doi.org/10.5281/zenodo.21434897)

Non-peer-reviewed preprint

Abstract

Background

Daily maximum temperature is associated with ground-level ozone, but a change in observed ozone between periods can reflect both a changed temperature distribution and a changed temperature-standardized ozone response. We asked how these two components differed between 2015–2019 and 2021–2025 across represented monitoring sites in the contiguous United States and District of Columbia.

Methods

We analyzed untransformed continuous maximum daily 8-hour average ozone (MDA8) concentrations from 884 balanced U.S. Environmental Protection Agency Air Quality System (EPA AQS) monitoring sites (2,396,553 site-days). Temperature came from matched NOAA/NCEI GHCN-Daily stations. Within each region's common temperature support, we fit a pooled block-diagonal, unregularized ordinary least-squares working model with an identity link, site-specific indicator terms, region-specific period intercepts, region-period natural-cubic temperature terms, and region-period cyclic seasonal terms. Fixed-calendar, equal-site standardization yielded a symmetric two-period decomposition. NOAA-region-stratified whole-site resampling provided 1,000-replicate percentile intervals.

Results

Nationally, the modeled later-minus-early ozone difference was +0.46 ppb (95% bootstrap percentile interval [0.35, 0.58]). The temperature-distribution component was +0.07 ppb ([0.04, 0.10]), and the temperature-standardized response component was +0.39 ppb ([0.28, 0.51]). The response component was larger in absolute magnitude nationally. Regional estimates were heterogeneous: total point estimates were positive in Northern Rockies and Plains, Northwest, Ohio Valley, South, Southwest, and Upper Midwest; negative in Northeast, Southeast, and West; the Northeast total interval included zero. Treating 2020 differently produced the largest descriptive departure among the sensitivity specifications: assigning 2020 to the later period yielded a total of +0.06 ppb ([−0.05, 0.16]). The broader-network, lower-complexity temperature spline, and event-provenance and annual-quality analyses were closer to the primary estimates: their national totals remained within 0.02 ppb of the primary estimate. In the descriptive elevated-ozone analysis, the site-equal proportion changed from 1.42% to 1.38%, a change of −0.04 percentage points ([−0.11, 0.03]).

Conclusions

Across represented monitoring sites, the later period had a positive modeled difference in continuous MDA8, with the temperature-standardized response component larger than the temperature-distribution component nationally. Regional patterns varied, and the estimate was sensitive to how 2020 was handled. The small negative descriptive elevated-day point estimate, whose interval included zero, concerns a different feature of the ozone distribution and is not a decomposition. The analysis is associational. Whole-site resampling preserved each selected site's observed time series, but the uncertainty procedure did not explicitly model serial correlation within sites or spatial dependence across sites.

Keywords

ground-level ozone; maximum daily 8-hour average ozone; temperature; monitoring sites; standardization; decomposition; whole-site bootstrap; regional heterogeneity

Introduction

Ground-level ozone varies with atmospheric chemistry, precursor availability, transport, deposition, and meteorological conditions. Temperature is associated with several ozone-conducive processes, and ozone–temperature relationships have been documented in observational and model-based work (Bloomer et al. 2009; Rasmussen et al. 2012; Pusede, Steiner, and Cohen 2015). These relationships are not expected to be spatially or temporally uniform. Studies have described geographic and recent temporal variation in U.S. ozone–temperature relationships and ozone trends (Ninneman and Jaffe 2021; Nolte et al. 2021; Li, Wang, and Lu 2025; Chang et al. 2025; Bao et al. 2025). The present study therefore treats its contribution as an incremental, reproducible update, not discovery of the basic association.

A later-minus-early ozone difference can arise through at least two statistical paths. First, the distribution of daily maximum temperature can shift while the fitted temperature–ozone response remains fixed. Second, the fitted ozone response at a standardized temperature distribution can differ between periods. A symmetric two-period decomposition separates these quantities without making one period the sole reference and without assigning either component to a specific physical or policy mechanism.

National aggregation can conceal regional variation in temperature ranges, ozone seasonality, monitoring networks, and fitted response patterns. We therefore estimated the decomposition within NOAA's framework of nine climatically consistent regions in the contiguous United States (NOAA National Centers for Environmental Information n.d.), followed by equal-site national aggregation. The represented sites define the target; they do not constitute a population-weighted sample of U.S. people or places.

The study objective was to decompose the modeled difference in expected MDA8 between 2015–2019 and 2021–2025 into a temperature-distribution component and a temperature-standardized response component, nationally and by region. The exact frozen hypotheses were:

1. In eastern NOAA climate regions, later-period temperature conditions may tend to increase expected MDA8 ozone, while the temperature-standardized ozone response at comparable temperatures will be lower.
2. In much of the eastern United States, the negative temperature-standardized response component is expected to outweigh any positive temperature-distribution component, producing lower expected MDA8 ozone overall.
3. Western regions are expected to show weaker, more heterogeneous, or potentially opposing response changes because background ozone, wildfire smoke, transport, terrain, and regional emissions sources may play larger roles.

These hypotheses are directional and associational. The frozen plan did not define exact East/West memberships, quantitative directional thresholds, point-versus-interval evidence, aggregation rules, sensitivity-disagreement handling, or a categorical decision vocabulary. They are therefore reported with descriptive evidence and the status **not formally adjudicable under the frozen plan**. This status is a reporting-plan limitation, not a scientific finding.

Methods

Study design and setting

This retrospective longitudinal analysis used public environmental-monitoring records for the contiguous United States and District of Columbia. The source window was January 1, 2015 through December 31, 2025. The primary comparison used equal-length early and later periods, 2015–2019 and 2021–2025. We excluded 2020 from the primary comparison to preserve two symmetric five-year windows and because it lies at the boundary between the defined periods; alternative assignments and a continuous-time specification assessed sensitivity to that boundary choice. February 29 was omitted to preserve a fixed non-leap seasonal calendar.

Ozone data and site-day construction

Hourly ozone observations came from the U.S. Environmental Protection Agency Air Quality System (EPA AQS)/AirData, using parameter 44201 and its documented bulk files (U.S. Environmental Protection Agency 2015, n.d.c). Maximum daily 8-hour average ozone (MDA8) values were reconstructed from rolling 8-hour averages. When multiple ozone monitors were collocated, a prespecified conservative selection rule was used. Parameter occurrence codes (POCs) containing duplicate records for a site-hour were excluded; the site-hour was retained only when exactly one eligible POC remained. Site-hours with multiple remaining POCs were excluded rather than averaged or selected by priority. Daily MDA8 construction followed the applicable EPA Appendix U conventions (U.S. Environmental Protection Agency n.d.a). Each candidate 8-hour window required at least 6 observed hourly values and averaged the 6–8 available values without substituting missing hours; a daily MDA8 value required at least 13 of 17 valid windows, with no concentration-dependent incomplete-day exception. Hourly concentrations and 8-hour means were truncated, not rounded, to three decimal ppm places before conversion to ppb (Supplement, Section S2.1). Primary data retained event-affected ambient observations; the event-provenance analysis applied its separately frozen row filter.

The primary outcome was continuous MDA8 in ppb without outcome transformation. The descriptive elevated-ozone analysis used a binary indicator defined as stored MDA8 after the specified truncation and before presentation rounding, strictly above 70.0 ppb. That indicator is a study-defined descriptive threshold, not a regulatory attainment or violation measure.

Temperature data and matching

Daily maximum temperature (TMAX) came from the National Oceanic and Atmospheric Administration's National Centers for Environmental Information (NOAA/NCEI) Global Historical Climatology Network–Daily (GHCN-Daily) (Menne, Durre, Korzeniewski, et al. 2012; Menne, Durre, Vose, et al. 2012). Quality-accepted TMAX records were converted to degrees Celsius and matched to EPA local-standard calendar dates. Each coordinate episode was matched independently to the nearest eligible station within 50 km with at least 90% coverage of that coordinate episode's observed in-season dates by quality-accepted TMAX records. A coordinate episode was a consecutive run of a site's observed in-season dates sharing the same EPA-reported latitude and longitude; different episodes from one site could therefore use different stations. The overlap denominator was the observed in-season ozone dates within the coordinate episode, and the numerator was the subset with quality-accepted TMAX. Distance and station identifier were deterministic tie breakers.

Eligibility, seasons, and analytical population

Using site, county, then state ozone-season precedence, a site-season-year qualified when days with valid reconstructed MDA8 and quality-accepted matched TMAX comprised at least 75% of all calendar days in its applicable official ozone season. No separate monitor-operational subset was used; February 29 counted when applicable. Primary sites required at least 4 qualifying years in each comparison period. The balanced primary population contained 884 sites, 2,396,553 site-days, 1,192,343 early rows, and 1,204,210 later rows. Eligibility was evaluated before common-support trimming and before February 29 was removed from model fitting and standardization.

Regions and common temperature support

Sites were assigned to 9 NOAA climate regions (NOAA National Centers for Environmental Information n.d.). The District of Columbia was assigned to the Northeast under the study's prespecified state-to-region crosswalk. Within each region, common support used 2 °C TMAX bins with at least 30 eligible site-days in each period. A region also required at least 20 sites. For each region and period, the retained fraction was the number of eligible balanced-site rows in retained common-support bins divided by all eligible balanced-site rows after excluding 2020 and before common-support or February 29 trimming; at least 80% had to be retained. Unsupported temperatures were not extrapolated.

Primary model

The primary specification was a pooled block-diagonal, unregularized ordinary least-squares working model with an identity link. Nine regional normal-equation solves were algebraically equivalent to the pooled block-diagonal fit and shared one pooled spline-basis state. Each regional block contained site indicators, a later-period intercept, and separate early/late four-column centered natural-cubic TMAX and six-column centered cyclic day-of-year blocks. Because sites did not cross regions and all other terms were region-specific, the blocks formed the direct-sum factorization of the pooled design; no coefficient was shared. The pooled support-trimmed TMAX state used boundaries [-21.9, 51.7] °C and internal knots [18.3, 25.6, 30.6] °C, calculated by the prespecified linear-interpolation quantile method implemented in NumPy. No regularization, outcome transformation, prediction clipping, or additional covariate selection was used.

Standardization and symmetric decomposition

For each site, the fixed represented calendar comprised the sorted distinct non-February-29 day-of-year values in its support-trimmed early or later rows; each represented day received equal weight. Period-specific empirical TMAX frequencies supplied temperature weights. Predictions were averaged within site over the Cartesian product of those temperatures and the fixed calendar before equal-site regional and national aggregation. This product construction treated the empirical temperature distribution independently of calendar day for the standardized prediction. The four standardized quantities were:

- A: early temperatures evaluated under the early fitted response;
- B: later temperatures evaluated under the early fitted response;
- C: early temperatures evaluated under the later fitted response; and
- D: later temperatures evaluated under the later fitted response.

The temperature-distribution component averaged the B-minus-A and D-minus-C contrasts. The temperature-standardized response component averaged the C-minus-A and D-minus-B contrasts. Their sum equaled the total D-minus-A difference within the prespecified numerical tolerance. Sites were equally weighted within regions and nationally; the national estimate was neither an unweighted regional mean nor a site-day-weighted quantity.

Bootstrap uncertainty

The primary bootstrap sampled complete sites with replacement independently within each NOAA region. Regional draw counts equaled primary site counts, duplicate site draws received distinct site labels, and each selected draw retained its complete early and later record. Replicate-specific support and spline state were rebuilt under the primary bootstrap specification. We targeted 1,000 successful replicates. Percentile intervals used the 2.5th and 97.5th empirical quantiles, calculated by linear interpolation using NumPy's `linear` method. No p-values or formal intervals for differences between specifications were calculated.

Sensitivity analyses

The sensitivity and robustness program included:

1. three alternative treatments of 2020: assigning it to the early period, assigning it to the later period, and a prospectively clarified continuous-time specification with a transition-year-only intercept and endpoint standardization;
2. a broader network requiring at least 1 qualifying year in each period, qualifying-site-year rows only, a common site set, and eligibility before support;
3. a lower-complexity temperature spline with exact pooled tertile-probability knots and otherwise unchanged model and population;
4. S4-A retained-only event-provenance filtering, S4-B stringent 2025 annual-quality filtering, and their intersection S4-C; and
5. the prespecified continuous-model diagnostic suite and a prospectively defined descriptive elevated-ozone analysis.

Every sensitivity retained the frozen outcome, model, support, basis, and weighting except where its documented scientific question explicitly changed one of those elements. The post-freeze amendment chronology is reported in the Supplement.

Descriptive elevated-ozone analysis

For each primary site and period, the descriptive elevated-day proportion was the number of retained days above 70.0 ppb divided by valid retained days. Sites with zero elevated days remained. Regional estimates averaged site-specific proportions within region; national estimates averaged all 884 site-specific proportions. The change was later minus early in percentage points. Secondary row-weighted counts and proportions were labeled separately.

The descriptive analysis used the same validated site-resampling draws as the primary bootstrap. Site-equal early, later, and percentage-point-change summaries were calculated in each replicate. It did not use binomial intervals, site-day resampling, a binary regression, or a threshold decomposition.

Diagnostics and rejected binary model

Primary diagnostics assessed design rank, solver status, coefficient and prediction finiteness, condition numbers, fitted range, residual quantiles, period and region-period summaries, residual variance by fitted-value decile, calibration, within-site lag-one residual correlation, and leverage diagnostics. The median within-site lag-one residual correlation was 0.54, indicating substantial temporal autocorrelation.

Whole-site resampling preserved each selected site's observed time series, but the uncertainty procedure did not explicitly model serial correlation within sites or spatial dependence across sites. Residual variance varied across fitted-value deciles, consistent with heteroskedasticity; the complete decile summaries appear in the Supplement.

The 884-site population included 55 sites with no elevated days, so a finite unregularized logistic maximum-likelihood estimate did not exist. Restricting the analysis to the 829 sites containing both elevated and non-elevated days still left quasi-complete separation in the Northwest. No penalized, conditional, Bayesian, Firth, bias-reduced, or other rescue model was fitted, and no binary coefficient or prediction is reported. The separate descriptive threshold analysis retained all 884 sites and did not fit a binary model.

Software, reproducibility, and prospective amendments

The analysis used the frozen Python environment and frozen machine-readable result records. Population, panel, configuration, code, bootstrap, and output identities were checksum-validated. Point fits and decompositions were repeated deterministically.

Continuous MDA8 was adopted as the primary outcome after the binary model was found to be non-estimable and before any substantive continuous-model result was examined. Several sensitivity definitions—including the exact S1-C time form, broader-network population, tertile-knot spline, Family 4 filters, and Family 5 descriptive estimand—were finalized after earlier analyses had been completed but before results from the corresponding analysis were examined. Bootstrap implementation rules for the 2020 family and Family 4 were finalized after point estimates but before the associated uncertainty intervals were calculated. The final hypothesis-reporting decision was made after all analyses and governs reporting only.

Results

Study population

The source panel yielded 884 balanced sites and 2,396,553 support-trimmed site-days: 1,192,343 in 2015–2019 and 1,204,210 in 2021–2025. All 9 regions remained estimable. Site and row distributions are summarized in Table 1 and Figure 1.

Table 1. Study population and analytical design

Category	Element	Primary specification
Population	Represented sites	884
Population	Site-days	2,396,553
Periods	Early / later rows	1,192,343 / 1,204,210

Table 1 (continued)

Category	Element	Primary specification
Periods	Comparison	2015–2019 versus 2021–2025; 2020 excluded
Geography	NOAA climate regions	9
Eligibility	Completeness	≥75% of official-season calendar days with valid MDA8 and quality-accepted matched TMAX; ≥4 qualifying years per period
Support	Regional common support	2 °C bins; ≥30 rows per period; ≥20 sites; ≥80% of eligible balanced-site region-period rows retained after 2020 exclusion and before February 29 removal; 234 retained bins
Model	Working model	Pooled block-diagonal, unregularized ordinary least-squares working model with an identity link
Model	Terms	site fixed effects; region-specific later-period intercepts; region-by-period four-column centered natural-cubic TMAX basis; region-by-period six-column centered cyclic day-of-year basis
Uncertainty	Bootstrap	1,000-replicate NOAA-region-stratified whole-site percentile bootstrap

Note: Rows are site-days. The completeness denominator is every calendar day in the applicable official ozone season; the numerator requires valid reconstructed MDA8 and quality-accepted matched TMAX. February 29 is included in eligibility when it lies in the official season, then excluded from fitting and standardization. The represented monitoring sites are not population-weighted exposure estimates. The primary comparison excludes 2020. The common-support retention denominator is all eligible balanced-site rows in each region-period after 2020 exclusion and before common-support or February 29 trimming.

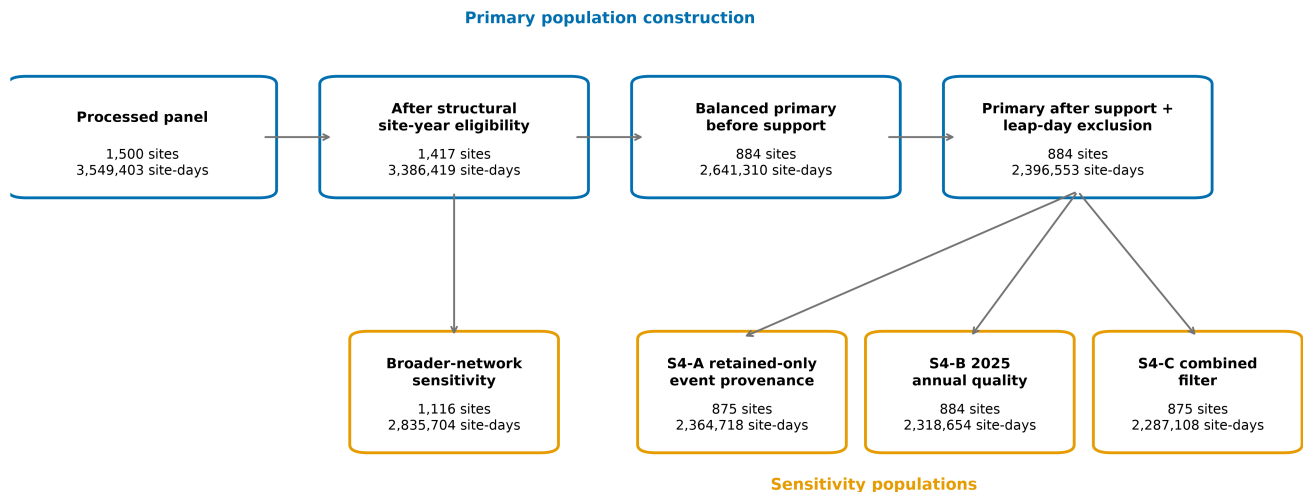


Figure 1. Monitoring-site analysis flow. The processed panel was restricted by frozen site-year completeness and balanced-site rules, regional common temperature support, and February 29 exclusion to the 884-site primary population. The broader-network and Family 4 populations are distinct sensitivity populations. Counts refer to represented monitoring sites and site-days, not people or a population-based sample.

Primary national decomposition

National A, B, C, and D were 42.07, 42.14, 42.45, and 42.53 ppb, respectively. The temperature-distribution component was +0.07 ppb ([0.04, 0.10]), and the temperature-standardized response component was +0.39 ppb ([0.28, 0.51]). The total modeled later-minus-early difference was +0.46 ppb ([0.35, 0.58]). Both components were positive and therefore reinforced each other; the response component was larger in absolute magnitude.

Regional heterogeneity

Regional estimates varied in direction and magnitude (Table 2; Figures 2 and 3). Total point estimates were positive in Northern Rockies and Plains, Northwest, Ohio Valley, South, Southwest, and Upper Midwest; negative in Northeast, Southeast, and West. Temperature and response components reinforced nationally and in eight regions, but opposed in the Ohio Valley. Total intervals were entirely above zero in Northern Rockies and Plains, Northwest, Ohio Valley, South, Southwest, and Upper Midwest; entirely below zero in Southeast and West; and included zero in Northeast. The Northwest response interval also included zero.

Table 2. Primary national and regional decomposition

Region	Sites	Temperature component, ppb [95% PI]	Response component, ppb [95% PI]	Total, ppb [95% PI]	Relation
National	884	+0.07 [0.04, 0.10]	+0.39 [0.28, 0.51]	+0.46 [0.35, 0.58]	reinforce
Northeast	158	-0.07 [-0.14, 0.01]	-0.10 [-0.29, 0.08]	-0.17 [-0.36, 0.02]	reinforce
Northern Rockies and Plains	34	+0.38 [0.27, 0.48]	+1.19 [0.63, 1.78]	+1.57 [0.98, 2.17]	reinforce
Northwest	21	+0.58 [0.33, 0.79]	+0.67 [-0.32, 1.69]	+1.25 [0.32, 2.17]	reinforce
Ohio Valley	133	-0.15 [-0.22, -0.07]	+0.47 [0.26, 0.68]	+0.32 [0.11, 0.54]	oppose
South	128	+0.61 [0.53, 0.68]	+1.31 [1.08, 1.55]	+1.92 [1.68, 2.18]	reinforce
Southeast	136	-0.15 [-0.24, -0.07]	-0.19 [-0.38, 0.02]	-0.34 [-0.57, -0.11]	reinforce
Southwest	96	+0.12 [0.08, 0.16]	+1.31 [0.95, 1.72]	+1.42 [1.05, 1.83]	reinforce
Upper Midwest	59	+0.36 [0.24, 0.50]	+1.18 [0.90, 1.47]	+1.54 [1.26, 1.84]	reinforce
West	119	-0.16 [-0.26, -0.05]	-0.79 [-1.32, -0.29]	-0.95 [-1.52, -0.43]	reinforce

Note: Site-equal estimates compare 2015–2019 with 2021–2025. PI denotes the empirical 95% whole-site bootstrap percentile interval. Relation compares the signs of the temperature and response component point estimates; 'reinforce' and 'oppose' are descriptive sign patterns, not significance tests or causal mechanisms. The response component is temperature-standardized and associational, not causal.

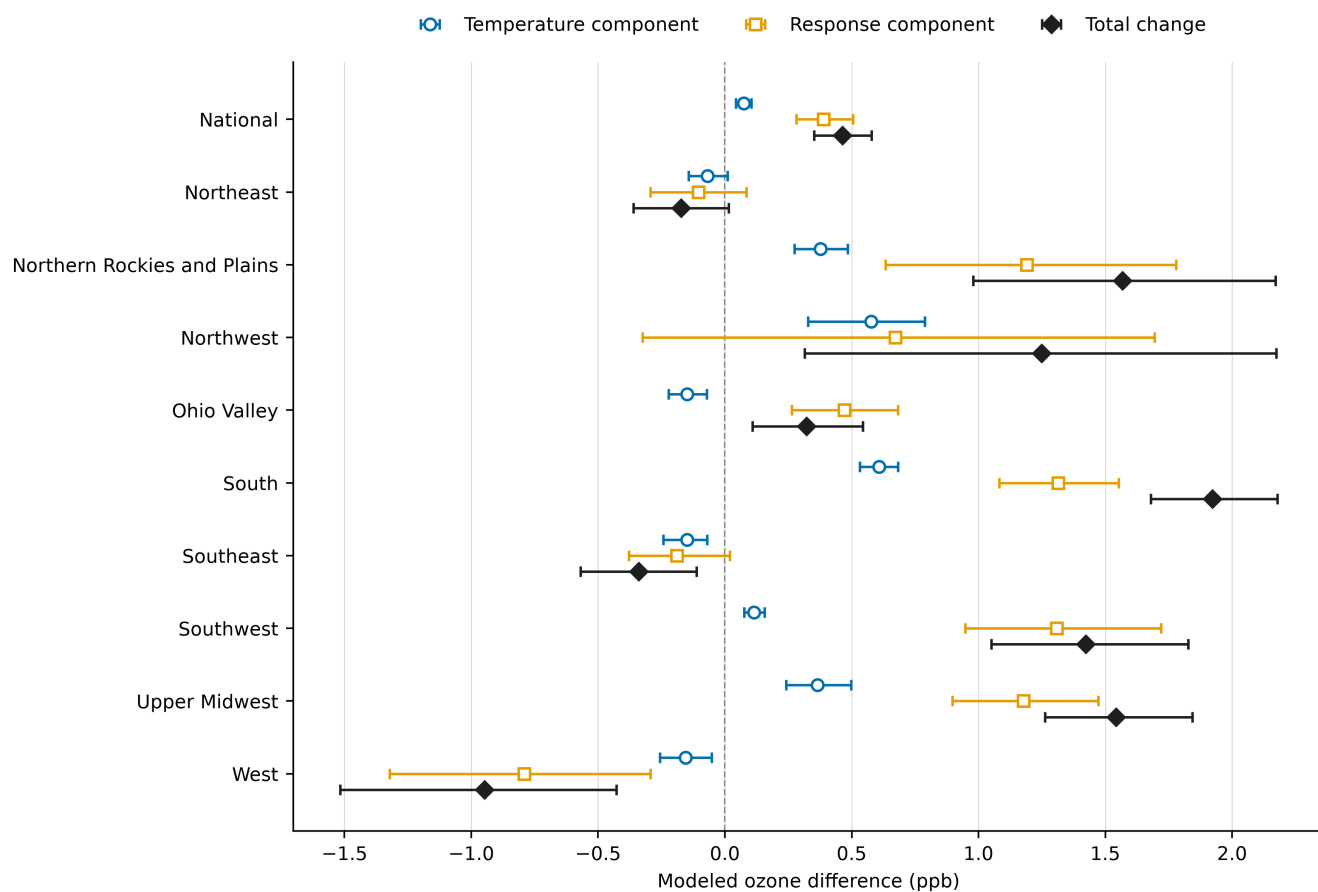


Figure 2. Primary national and regional decomposition. Points show site-equal modeled ozone differences and horizontal bars show empirical 95% NOAA-region-stratified whole-site bootstrap percentile intervals. The response component is temperature-standardized and associational; the decomposition does not identify causal mechanisms.

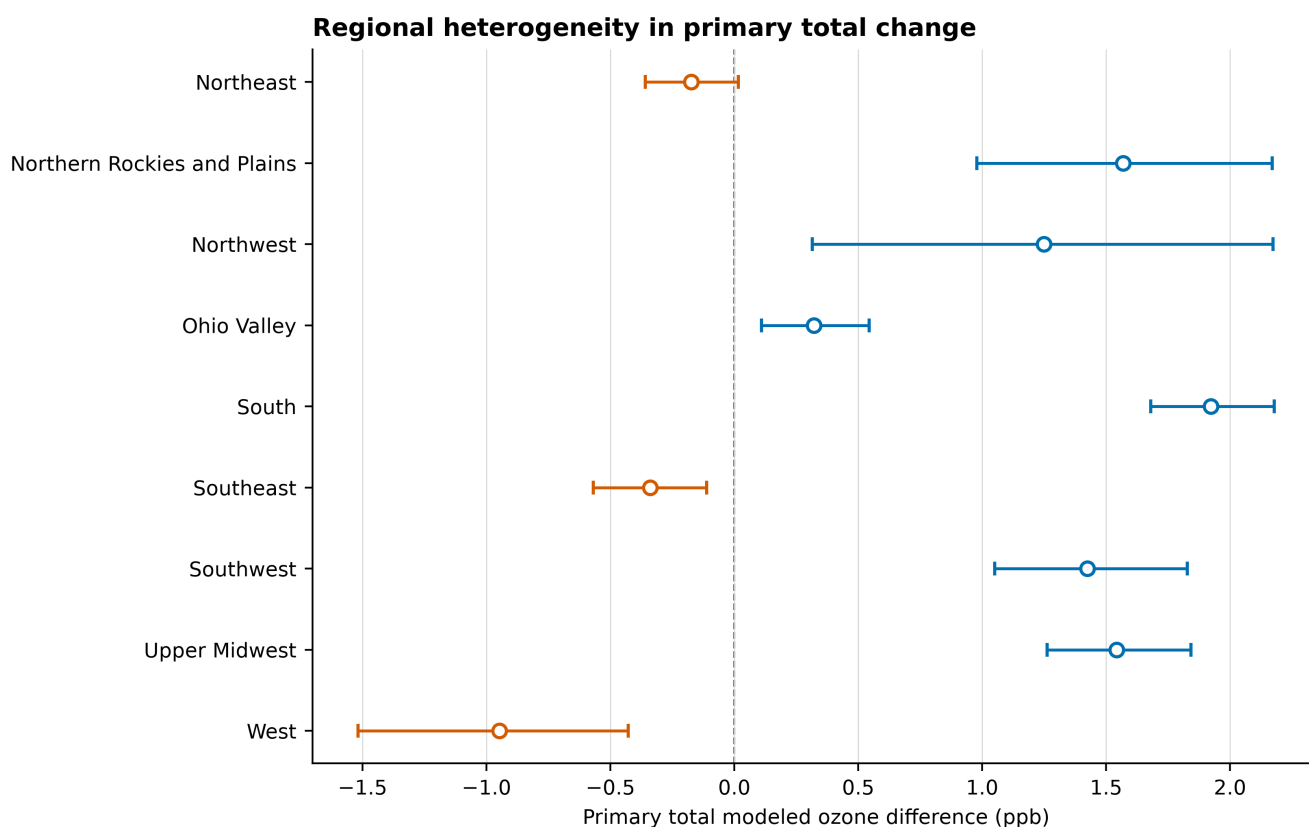


Figure 3. Regional heterogeneity in the primary total modeled ozone difference. Horizontal bars are empirical 95% region-stratified whole-site bootstrap percentile intervals. Blue points denote positive point estimates and vermilion points denote negative point estimates; interval position, rather than color alone, determines relation to zero.

Bootstrap uncertainty

The primary bootstrap obtained 1,000 successful replicates, all of which satisfied the prespecified population, support, rank, decomposition-identity, and finite-value checks. The national and regional percentile intervals are shown in Table 2.

Sensitivity analyses

Table 3 and Figure 4 compare every continuous sensitivity with the primary analysis. Across all national and regional component and total comparisons relative to the primary analysis, there were seven sign changes, 23 interval-relation changes, four component-relation changes, and 16 absolute point differences of at least 0.5 ppb. All sign changes, component-relation changes, and absolute point differences of at least 0.5 ppb occurred within the 2020 family. Interval-relation changes occurred in S1-A (six), S1-B (ten), S1-C (two), the broader-network analysis (four), and S4-C (one); the S4-C change was the Southeast response component, whose interval changed from including zero under the primary analysis to entirely below zero. The specification assigning 2020 to the later period produced the largest descriptive departure among the sensitivity specifications: its national response component was +0.01 ppb ([-0.10, 0.12]) and its total was +0.06 ppb ([-0.05, 0.16]). Together, these results show that handling of 2020 is a material sensitivity of the national estimate.

The broader-network result used 1,116 sites and was descriptively close to primary nationally, with the same national signs and component relation. The lower-complexity spline result was nearly identical to primary nationally and regionally. S4-A, S4-B, and S4-C were descriptively close to primary, with unchanged national signs, interval relations, and component relation. Regional sign, interval-relation, and reinforce/oppose changes are reported completely in the Supplement; no formal between-specification inference was performed.

Table 3. Sensitivity-analysis summary

Specification	Population	Estimand (units)	Point	95% PI	Difference from primary	Same sign as primary?	Same interval relation as primary?	Same component relation as primary?
Primary (2020 excluded)	884 sites; 2,396,553 fit rows	continuous total change (ppb)	+0.463	[0.352, 0.579]	—	—	—	—
S1-A (2020 assigned early)	952 sites; 2,788,753 fit rows	continuous total change (ppb)	+0.829	[0.728, 0.926]	+0.366	Yes	Yes	Yes
S1-B (2020 assigned later)	936 sites; 2,773,587 fit rows	continuous total change (ppb)	+0.062	[-0.050, 0.164]	-0.401	Yes	No	Yes
S1-C (continuous-time specification)	884 sites; 2,638,658 fit rows	continuous total change (ppb)	+0.825	[0.657, 0.989]	+0.361	Yes	Yes	Yes
Broader eligible network	1,116 sites; 2,835,704 fit rows	continuous total change (ppb)	+0.444	[0.333, 0.549]	-0.019	Yes	Yes	Yes
Three-degree-of-freedom TMAX spline	884 sites; 2,396,553 fit rows	continuous total change (ppb)	+0.463	[0.352, 0.578]	-0.0005	Yes	Yes	Yes
S4-A (retained-only event-provenance filter)	875 sites; 2,364,718 fit rows	continuous total change (ppb)	+0.467	[0.361, 0.573]	+0.003	Yes	Yes	Yes
S4-B (stringent 2025 annual quality)	884 sites; 2,318,654 fit rows	continuous total change (ppb)	+0.471	[0.359, 0.586]	+0.007	Yes	Yes	Yes
S4-C (event and 2025 quality combined)	875 sites; 2,287,108 fit rows	continuous total change (ppb)	+0.473	[0.364, 0.581]	+0.010	Yes	Yes	Yes
Descriptive elevated-ozone analysis	884 sites; 2,396,553 rows	later-minus-early change (percentage points)	-0.037	[-0.115, 0.034]	Not comparable	Not comparable	Not comparable	Not comparable

Note: The 'Difference from primary' column is descriptive; no interval or p-value was calculated for between-specification differences. The nonzero three-degree-of-freedom spline difference that would round to zero at three decimals is shown to four decimals to preserve its sign. Interval relation identifies whether the 95% bootstrap percentile interval is above zero, below zero, or includes zero; component relation identifies whether the temperature and response components reinforce or oppose. The threshold row uses percentage points and must not be compared quantitatively with ppb. All continuous intervals are whole-site bootstrap percentile intervals.

National continuous estimates across primary and sensitivity specifications

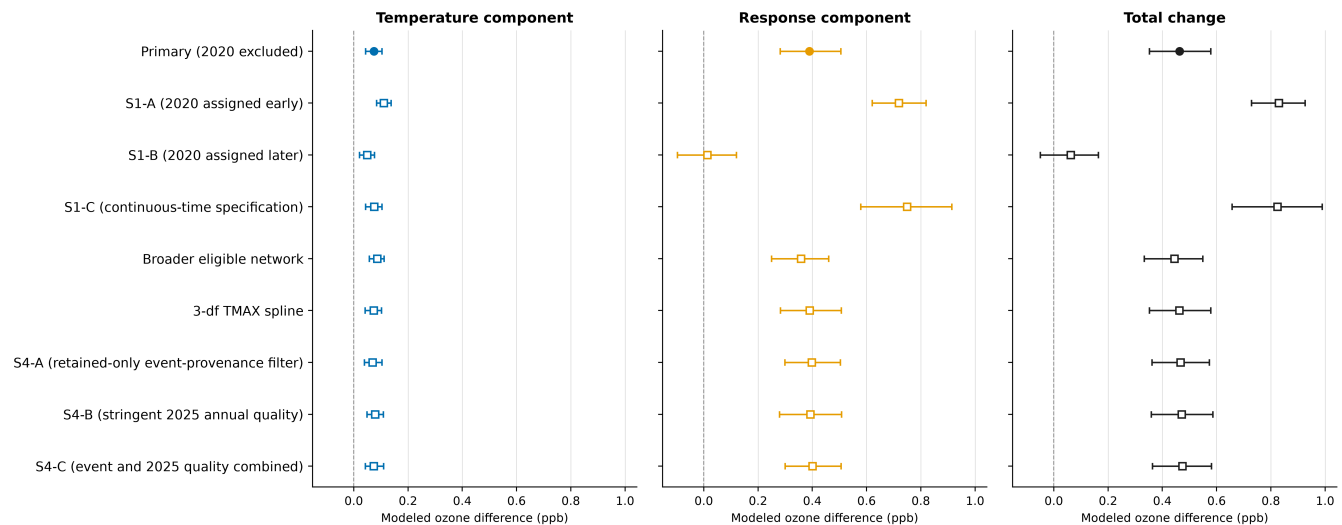


Figure 4. National estimates across the primary and continuous sensitivity specifications. Points show site-equal estimates and horizontal bars show empirical 95% whole-site bootstrap percentile intervals. Panels share the same ppb axis. The compact label “3-df TMAX spline” denotes the three-degree-of-freedom TMAX spline. Differences across specifications are descriptive; no formal between-specification inference was performed.

Descriptive elevated-ozone results

The national equal-site elevated-day percentage was 1.42% in the early period and 1.38% in the later period. The later-minus-early change was -0.04 percentage points ([-0.11, 0.03]), and its interval included zero. Change intervals were entirely above zero in Northern Rockies and Plains, Ohio Valley, South, Southwest, and Upper Midwest; entirely below zero in Northeast, Southeast, and West; and included zero in Northwest. The national descriptive point estimate was negative and differed in sign from the positive continuous-total point estimate, but the two quantities measure different distributional features and are not competing estimates of one estimand.

Table 4. Descriptive elevated-ozone results

Region	Sites	Early, % [95% PI]	Later, % [95% PI]	Change, percentage points [95% PI]	Point-estimate sign vs continuous total
National	884	1.42 [1.24, 1.60]	1.38 [1.20, 1.56]	-0.04 [-0.11, 0.03]	differing
Northeast	158	1.16 [0.98, 1.35]	0.73 [0.60, 0.88]	-0.43 [-0.52, -0.34]	concordant
Northern Rockies and Plains	34	0.13 [0.06, 0.22]	0.55 [0.43, 0.70]	+0.42 [0.28, 0.57]	concordant
Northwest	21	0.90 [0.57, 1.29]	0.61 [0.25, 1.01]	-0.29 [-0.58, 0.05]	differing
Ohio Valley	133	0.71 [0.59, 0.85]	0.82 [0.71, 0.96]	+0.12 [0.03, 0.21]	concordant
South	128	0.76 [0.62, 0.92]	1.23 [1.00, 1.50]	+0.47 [0.35, 0.60]	concordant
Southeast	136	0.28 [0.21, 0.37]	0.17 [0.13, 0.23]	-0.11 [-0.16, -0.07]	concordant
Southwest	96	1.32 [1.06, 1.59]	2.02 [1.65, 2.40]	+0.70 [0.45, 0.94]	concordant
Upper Midwest	59	0.85 [0.61, 1.14]	1.28 [1.03, 1.57]	+0.43 [0.27, 0.59]	concordant
West	119	5.39 [4.17, 6.76]	4.33 [3.11, 5.60]	-1.06 [-1.56, -0.64]	concordant

Note: Elevated ozone is stored MDA8 after the specified truncation and before presentation rounding, strictly above 70.0 ppb. Percentages are equal-site means, not pooled site-day percentages. ‘Concordant’ means the threshold-change and continuous-total point estimates have the same sign; ‘differing’ means their signs differ. The descriptive change is not a decomposition and uses unlike units from the continuous result.

Continuous and descriptive threshold results measure different distributional features

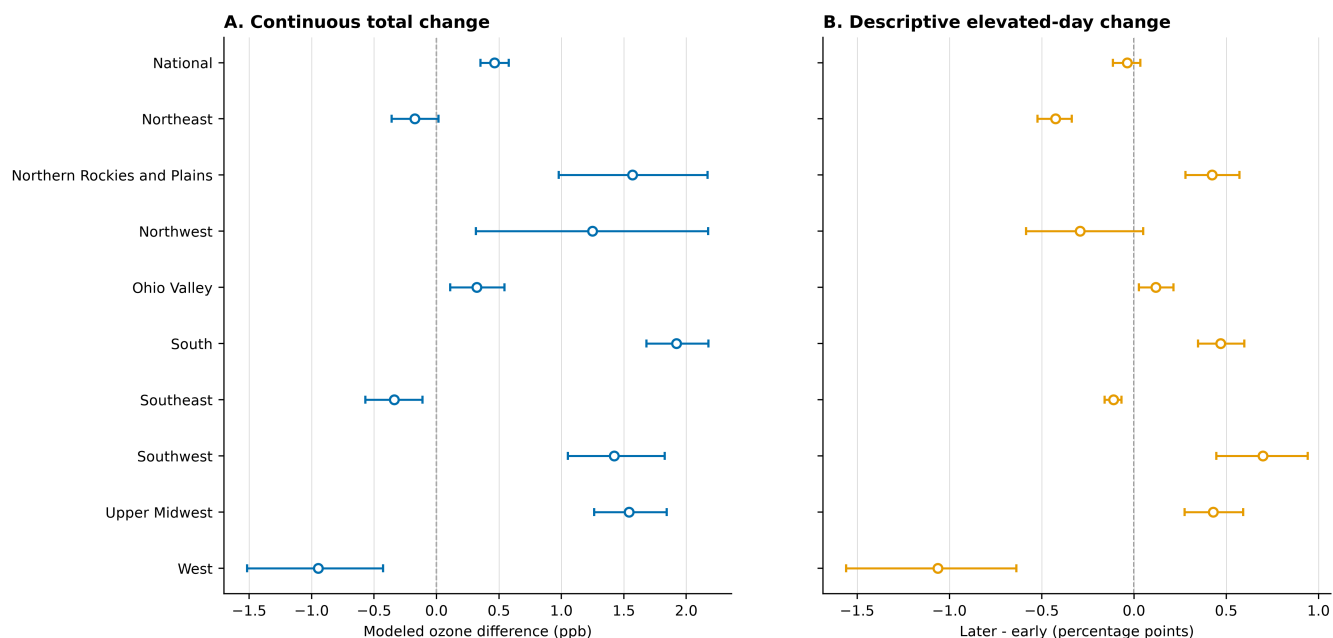


Figure 5. Primary continuous total modeled ozone differences (panel A, ppb) and descriptive changes in the equal-site proportion of days with stored MDA8 after the specified truncation and before presentation rounding, strictly above 70.0 ppb (panel B, percentage points). Points and bars are estimates and empirical 95% whole-site bootstrap percentile intervals. Separate axes prevent quantitative comparison of unlike units; the threshold result is not a decomposition and is not directly comparable with the continuous estimand.

Model diagnostics and binary-model failure

All nine regional design matrices were full rank, all solvers completed successfully, and all coefficients and predictions were finite. Across the nine regional blocks, the direct-sum design contained 1,073 columns and had summed rank 1,073. The root mean square error (RMSE) was 8.99 ppb; fitted values ranged from 15.15 to 80.57 ppb, and none was below zero or above the observed maximum. The median within-site lag-one residual correlation of 0.54 indicated substantial temporal autocorrelation.

The 884-site population included 55 sites with no elevated days, and a finite unregularized logistic maximum-likelihood estimate did not exist. Restriction to the 829 sites containing both elevated and non-elevated days still left quasi-complete separation in the Northwest. No penalized, conditional, Bayesian, Firth, bias-reduced, or other rescue model was fitted. This model failure was distinct from the descriptive threshold analysis.

Discussion

Principal findings

Across represented monitoring sites, the primary continuous analysis estimated a positive national modeled MDA8 difference between the later and early periods. The temperature-standardized response component was larger in absolute magnitude than the temperature-distribution component nationally. Regional patterns were heterogeneous, with positive total point estimates in several interior and southern regions, negative total point estimates in the West and Southeast, and a Northeast interval that included zero. Alternative handling of 2020 materially changed the national response and total, whereas broader-network, lower-complexity spline, and event-provenance and annual-quality analyses were descriptively closer to the primary result.

The descriptive elevated-day analysis produced a small negative national point estimate whose interval included zero. This does not contradict the positive difference in continuous MDA8: an equal-site modeled mean difference and the frequency above a fixed upper threshold describe different features of the ozone distribution. The descriptive threshold result was not decomposed into temperature and response components.

Interpretation of the response component

The response component describes how the fitted ozone response differed between periods after standardizing temperature distributions and calendar weights. It is not causal and cannot distinguish precursor emissions, smoke, humidity, wind, transport, deposition, monitoring changes, or other unmeasured factors. The response component's larger absolute magnitude nationally is a statistical feature of the fitted response, not evidence of a mechanism or policy change.

Regional heterogeneity

The largest positive primary total point estimates occurred in the South, Northern Rockies and Plains, Upper Midwest, and Southwest. The West and Southeast had negative total point estimates. The Northeast total and both components had intervals including zero; the Northwest response interval also included zero. In the Ohio Valley, a negative temperature component opposed a larger positive response component. Regional estimates should be read within their represented monitoring-site populations and supported temperature ranges. Differences among regions may reflect multiple measured and unmeasured conditions; the analysis does not identify a specific explanation.

Why transition-year handling matters

Assigning 2020 to the later period reduced the national response component from +0.39 to +0.01 ppb and the total from +0.46 to +0.06 ppb; both S1-B intervals included zero. The 2020 sensitivity family was prespecified to assess exclusion or assignment of the boundary year to a comparison period; the exact S1-C continuous-time form was prospectively clarified under the documented amendment chronology before the corresponding sensitivity result was available. The observed variation across these specifications limits any interpretation that depends on one period boundary. No interval or p-value was calculated for between-specification differences.

Other sensitivity analyses

National totals for the broader network, three-degree-of-freedom TMAX spline, and all three Family 4 filters differed from primary by less than 0.02 ppb, although regional differences are reported in full. The broader network examined dependence on the primary four-qualifying-year restriction. The lower-complexity spline changed only TMAX functional-form complexity. Family 4 changed event-provenance or annual-quality row inclusion while retaining the primary support and basis. Despite their numerical proximity to the primary analysis, these sensitivities addressed distinct population, functional-form, and data-quality questions.

Relationship to previous work

The temperature–ozone association and its regional variability are established topics (Bloomer et al. 2009; Rasmussen et al. 2012; Pusede, Steiner, and Cohen 2015; Ninneman and Jaffe 2021; Nolte et al. 2021). Recent U.S. work also examines changing ozone–temperature relationships and ozone trends (Li, Wang, and Lu 2025; Chang et al. 2025; Bao et al. 2025). Wells et al. and Camalier et al. estimated ozone trends adjusted for interannual meteorological variability (Wells et al. 2021; Camalier, Cox, and Dolwick 2007). The present study instead decomposes a two-period modeled difference into a temperature-distribution component and a temperature-standardized response component. Its contribution is a transparent represented-site decomposition through 2025, with complete reporting of the specified and prospectively amended sensitivity analyses and diagnostics. The targeted literature review was not systematic and cannot establish priority or superiority.

Strengths and limitations

Strengths include immutable official-source snapshots, an auditable site-day construction, outcome-independent matching and completeness rules, balanced cross-period sites, explicit common temperature support, equal-site standardization, a symmetric decomposition, whole-site resampling, complete reporting of all specified and prospectively amended sensitivity analyses, and deterministic reproduction checks.

Several limitations are central. First, the design is observational and associational; the response component is not causal. Second, represented monitoring sites are not a population-weighted measure of people, places, or personal exposure.

Third, residuals showed substantial within-site temporal autocorrelation. Whole-site resampling preserved each selected site's observed time series, but the uncertainty procedure did not explicitly model serial correlation within sites or spatial dependence across sites.

Fourth, residual variance varied across fitted-value deciles, consistent with heteroskedasticity. Fifth, the empirical site network changed over time, and the primary balanced-network restriction trades broader coverage for cross-period comparability. Sixth, conclusions were sensitive to handling of 2020. Seventh, event provenance and 2025 annual completeness/certification came from source fields whose interpretation and timing required prospective amendments. EPA's CY2025 guidance states that certification reflects agency submission and validation, gives a May 1, 2026 certification deadline, notes that data may remain under review and change before that deadline, advises outside users to exercise caution during that period, and distinguishes certification from EPA evaluation flags (U.S. Environmental Protection Agency 2026). The Family 4 rules were prospectively frozen study filters, not EPA-recommended analytical inclusion rules.

Eighth, the unregularized identity-link ordinary least-squares working model does not constrain predictions to be nonnegative; fitted values in this analysis nevertheless remained within the observed outcome range. Diagnostics were disclosed rather than used to select another estimator. Ninth, the descriptive threshold analysis measures a different feature of the ozone distribution and is not a substitute for the continuous decomposition. Tenth, the original binary site-indicator model was structurally non-estimable. The outcome-dependent 829-site restriction was examined only as an estimability check and still showed quasi-complete separation in the Northwest; no fitted outcome-selected, penalized, conditional, Bayesian, Firth, bias-reduced, or other rescue model was reported. Eleventh, humidity, wind, transport, precursor conditions, smoke, and other meteorological or emissions-related factors were not directly modeled. The analysis makes no direct emissions, smoke, regulatory, policy, or population-exposure attribution.

Finally, several definitions were prospectively amended after earlier results were available, although each family-specific rule was fixed before results from the corresponding sensitivity analysis were examined. Bootstrap implementation rules were finalized after point estimates but before the associated uncertainty intervals were calculated. Hypothesis reporting was resolved only after all analyses: because the frozen plan lacked a unique decision rule, all hypotheses remain **not formally adjudicable under the frozen plan**.

Conclusion

The primary continuous analysis found a positive site-equal modeled MDA8 difference between 2015–2019 and 2021–2025 across represented U.S. monitoring sites, with a larger national temperature-standardized response component than the temperature-distribution component and substantial regional heterogeneity. The result was especially sensitive to handling of 2020, while the broader-network, lower-complexity spline, and event-provenance and annual-quality analyses were descriptively closer to primary. The descriptive elevated-day point estimate was in the opposite direction nationally, although its interval included zero; that distinct estimand does not estimate the same quantity as the continuous decomposition. Findings are associational and should be interpreted within the represented monitoring network and limitations related to temporal and spatial dependence.

Data availability

The analysis uses public EPA AQS/AirData and NOAA/NCEI GHCN-Daily records (U.S. Environmental Protection Agency 2015, n.d.c; Menne, Durre, Korzeniewski, et al. 2012). EPA states that AQS ambient monitoring data are public domain and may be downloaded and used without requesting permission (U.S. Environmental Protection Agency n.d.b). Raw-source URLs, retrieval metadata, byte sizes, and checksums are documented in repository manifests. The processed analytical panel is not distributed with this version. Its possible future release remains subject to a separate redistribution review.

Code availability

Curated code and reproducibility materials are publicly available at <https://github.com/vardenellis/temperature-ozone-us-2015-2025>. The processed analytical panel is not included, and full reconstruction requires the official EPA and NOAA source records under the frozen acquisition rules. No separate software DOI has yet been issued.

Ethics statement

This study used public environmental-monitoring records, enrolled no human participants, used no individual-level personal information, and required no human-subject intervention or consent.

Funding

This research received no external funding.

Conflicts of interest

The author declares no competing interests.

Author contributions

Ellis Varden: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing—original draft, Writing—review and editing, Visualization, and Project administration.

AI assistance

AI-assisted tools contributed to planning, code scaffolding, source discovery, draft editing, and reporting automation. The author remains responsible for source verification, methodological decisions, execution, interpretation, and final content. No numerical finding is included without a machine-readable artifact and traceability record.

References

- Bao, Jiemeng, Xin Li, Liuwei Kong, Jie Li, Qi Chen, and Yuanhang Zhang. 2025. "Comparative Analysis of the Impact of Rising Temperatures on Ozone Levels in China and the United States." *npj Clean Air* 1: 23. <https://doi.org/10.1038/s44407-025-00023-8>.
- Bloomer, Bryan J., Jeffrey W. Stehr, Charles A. Piety, Ross J. Salawitch, and Russell R. Dickerson. 2009. "Observed Relationships of Ozone Air Pollution with Temperature and Emissions." *Geophysical Research Letters* 36 (9): L09803. <https://doi.org/10.1029/2009GL037308>.
- Camalier, Louise, William Cox, and Pat Dolwick. 2007. "The Effects of Meteorology on Ozone in Urban Areas and Their Use in Assessing Ozone Trends." *Atmospheric Environment* 41 (33): 7127–37. <https://doi.org/10.1016/j.atmosenv.2007.04.061>.
- Chang, Kai-Lan, Brian C. McDonald, Colin Harkins, and Owen R. Cooper. 2025. "Surface Ozone Trend Variability Across the United States and the Impact of Heat Waves (1990–2023)." *Atmospheric Chemistry and Physics* 25: 5101–32. <https://doi.org/10.5194/acp-25-5101-2025>.
- Li, Shuai, Haolin Wang, and Xiao Lu. 2025. "Anthropogenic Emission Controls Reduce Summertime Ozone–Temperature Sensitivity in the United States." *Atmospheric Chemistry and Physics* 25: 2725–43. <https://doi.org/10.5194/acp-25-2725-2025>.
- Menne, Matthew J., Imke Durre, Bryant Korzeniewski, Shelley McNeill, Kristy Thomas, Xungang Yin, Steven Anthony, et al. 2012. "Global Historical Climatology Network - Daily (GHCN-Daily), Version 3." NOAA National Climatic Data Center. <https://doi.org/10.7289/V5D21VHZ>.
- Menne, Matthew J., Imke Durre, Russell S. Vose, Byron E. Gleason, and Tamara G. Houston. 2012. "An Overview of the Global Historical Climatology Network-Daily Database." *Journal of Atmospheric and Oceanic Technology* 29 (7): 897–910. <https://doi.org/10.1175/JTECH-D-11-00103.1>.
- Ninneman, Madeline, and Daniel A. Jaffe. 2021. "Observed Relationship Between Ozone and Temperature for Urban Nonattainment Areas in the United States." *Atmosphere* 12 (10): 1235. <https://doi.org/10.3390/atmos12101235>.
- NOAA National Centers for Environmental Information. n.d. "U.S. Climate Regions." Accessed July 18, 2026. <https://www.ncei.noaa.gov/access/monitoring/reference-maps/us-climate-regions>.
- Nolte, Christopher G., Tiffany L. Spero, Jared H. Bowden, Marcus C. Sarofim, Jeremy Martinich, and Michael S. Mallard. 2021. "Regional Temperature-Ozone Relationships Across the U.S. Under Multiple Climate and Emissions Scenarios." *Journal of the Air & Waste Management Association* 71 (10): 1251–64. <https://doi.org/10.1080/10962247.2021.1970048>.
- Pusede, Sally E., Allison L. Steiner, and Ronald C. Cohen. 2015. "Temperature and Recent Trends in the Chemistry of Continental Surface Ozone." *Chemical Reviews* 115 (10): 3898–918. <https://doi.org/10.1021/cr5006815>.
- Rasmussen, D. J., A. M. Fiore, V. Naik, L. W. Horowitz, S. J. McGinnis, and M. G. Schultz. 2012. "Surface Ozone-Temperature Relationships in the Eastern US: A Monthly Climatology for Evaluating Chemistry-Climate Models." *Atmospheric Environment* 47: 142–53. <https://doi.org/10.1016/j.atmosenv.2011.11.021>.
- U.S. Environmental Protection Agency. 2015. "AirData Download Files Documentation." December 1, 2015. <https://aqs.epa.gov/aqsweb/airdata/FileFormats.html>.
- U.S. Environmental Protection Agency. 2026. "Questions and Answers on Ambient Air Monitoring Data Certification for CY2025 Data." February 2026. https://www.epa.gov/system/files/documents/2026-02/ambient-air-monitoring-data-certification-qa-for-cy2025_final.pdf.

References (continued)

- U.S. Environmental Protection Agency. n.d.a. "40 CFR Part 50, Appendix U: Interpretation of the Primary and Secondary National Ambient Air Quality Standards for Ozone." Electronic Code of Federal Regulations. Accessed July 14, 2026. <https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-50/appendix-Appendix%20U%20to%20Part%2050>.
- U.S. Environmental Protection Agency. n.d.b. "Do I Need to Request Permission to Use Monitoring Data and Graphics from AirData?" Accessed July 18, 2026. <https://www.epa.gov/outdoor-air-quality-data/do-i-need-request-permission-use-monitoring-data-and-graphics-airdata>.
- U.S. Environmental Protection Agency. n.d.c. "Pre-Generated Data Files." Accessed July 14, 2026. https://aqs.epa.gov/aqsweb/airdata/download_files.html.
- Wells, Benjamin, Pat Dolwick, Brian Eder, Mark Evangelista, Kristen Foley, Elizabeth Mannshardt, Chris Misenis, and Anthony Weishampel. 2021. "Improved Estimation of Trends in U.S. Ozone Concentrations Adjusted for Interannual Variability in Meteorological Conditions." *Atmospheric Environment* 248: 118234. <https://doi.org/10.1016/j.atmosenv.2021.118234>.